



R(earch) F(unding) K(nowledge) G(raph)

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Fernando Pérez Alonso-Castrillo



Licencia y referencias

Software:

Este proyecto se distribuye bajo la licencia Apache 2.0
(<https://www.apache.org/licenses/LICENSE-2.0>)

Referencias:

Material teórico basado en las diapositivas de la asignatura *Open Science and Artificial Intelligence in Research Software Engineering* (Esteban González y Daniel Garijo).



source: <https://www.shutterstock.com>



Research
Funding
Knowledge
Graph

RFKG

Connecting research. Funding science. Building knowledge.

Imagen generada con IA generativa
([ChatGPT.com](https://chatgpt.com))

Funding

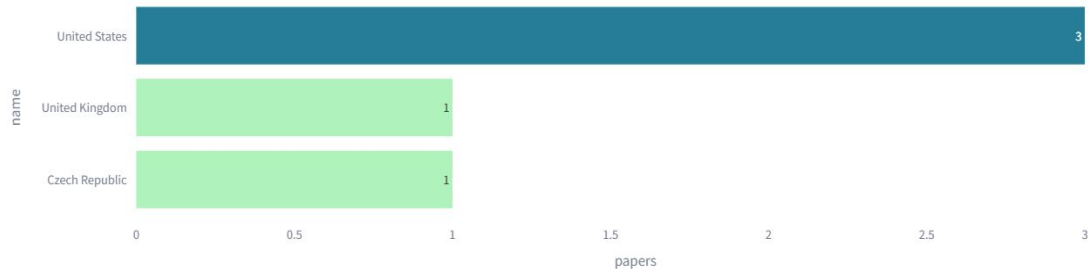
Vista principal: rankings de financiación filtrables por área temática.

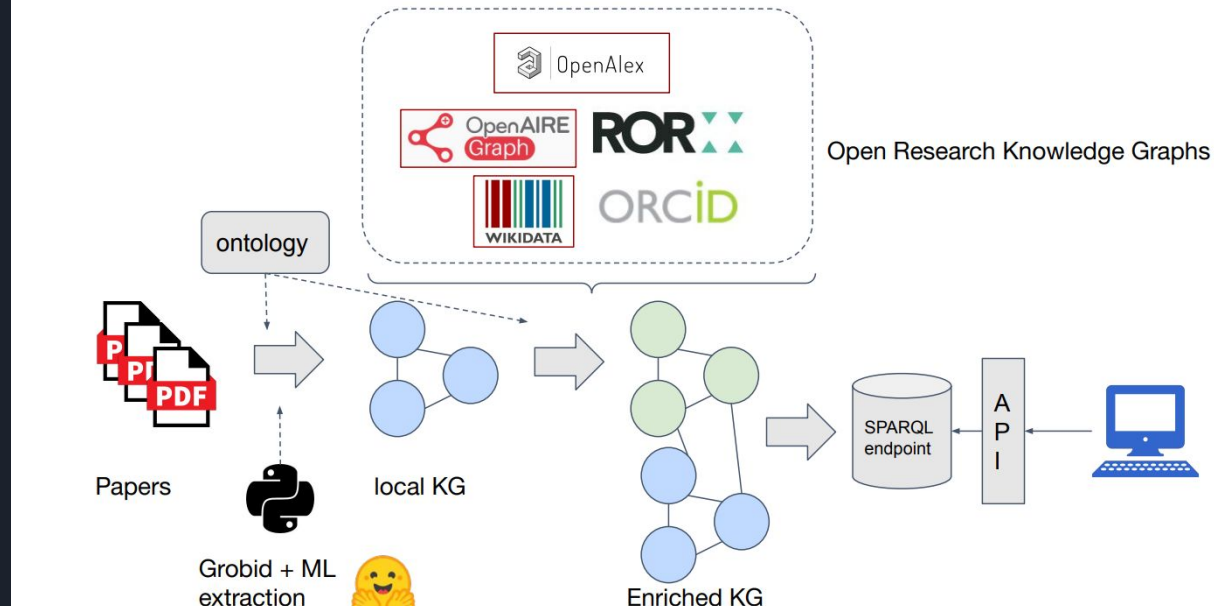
Filtrar rankings por topic

Todos

Ranking de países

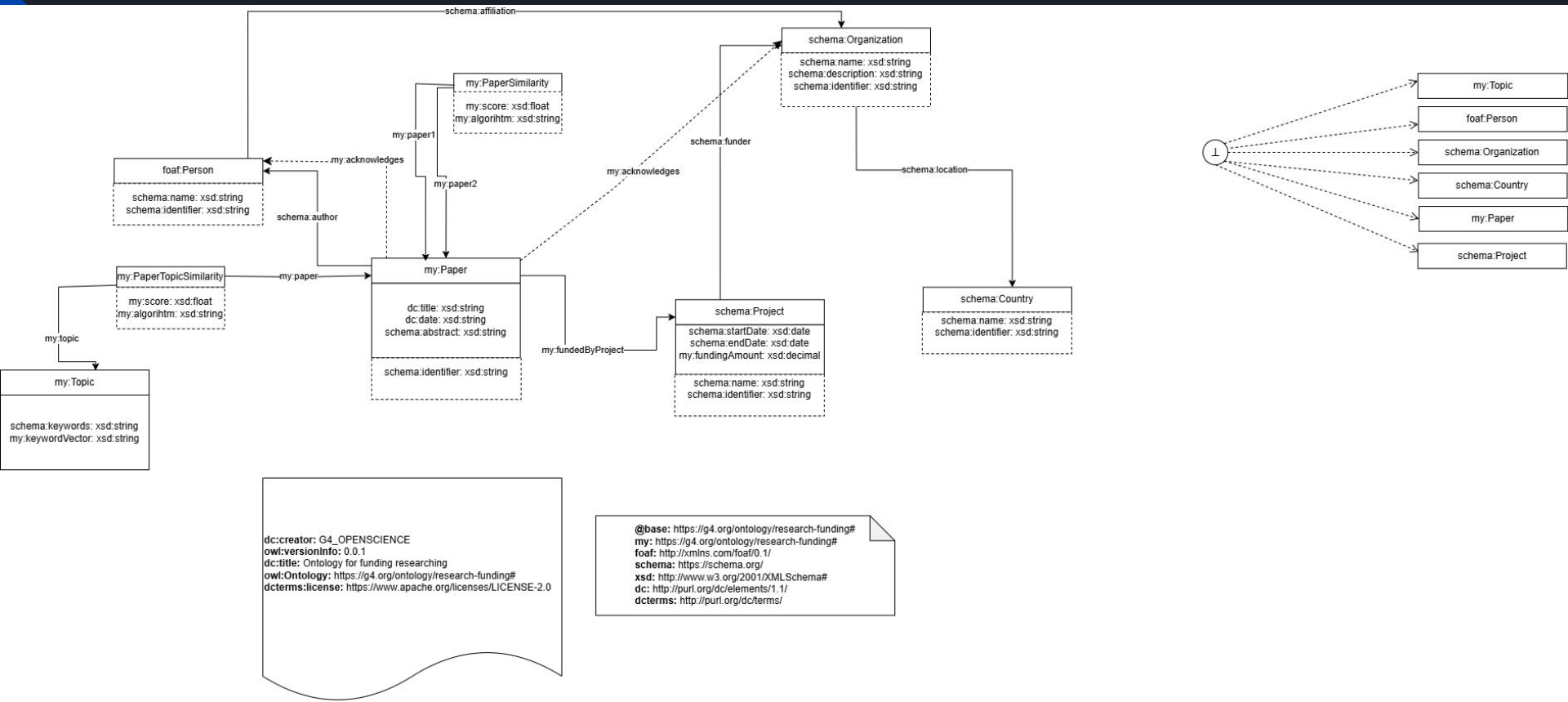
Top 5 países por papers financiados





source: <http://www.oeg-upm.net/>





Context, Reasoning, and Hierarchy: A Cost-Performance Study of Compound LLM Agent Design in an Adversarial POMDP

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Abstract

Deploying compound LLM agents in adversarial, partially observable sequential environments requires navigating several interacting design dimensions: (1) what the agent sees, (2) how it reasons, and (3) how tasks are decomposed across components. Yet practitioners lack guidance on which design choices improve performance versus merely increase inference costs. We present a controlled study of compound LLM agent design in CybORG CAGE-2, a cyber defense environment modeled as a Partially Observable Markov Decision Process (POMDP). Reward is non-positive, so all configurations operate in a failure-mitigation mode and errors compound over time. Our evaluation spans five model families, six models, and twelve configurations (3,475 episodes) with token-level cost accounting. We systematically vary context representation (raw observations vs. a deterministic, programmatic environment state-tracking layer with compressed history), deliberation (self-questioning, self-critique, and self-improvement tools, with optional chain-of-thought prompting), and hierarchical decomposition (monolithic ReAct vs. delegation to specialized sub-agents). We find that: (1) Programmatic state abstraction delivers the largest returns per token spent (RPTS), improving mean return by up to 76% over raw observations. (2) Distributing deliberation tools across a hierarchy degrades performance relative to hierarchy alone for all five model families, reaching up to 3.4x worse mean return while using 1.8-2.7x more tokens. We call this destructive interaction pattern a *deliberation cascade*. (3) Hierarchical decomposition without deliberation tools achieves the best absolute performance for most models, and context engineering is generally more cost-effective than deliberation. These findings suggest a design principle for structured adversarial POMDPs: invest in programmatic infrastructure.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; **Partially-observable Markov decision processes**; **Multi-agent systems**; **Intelligent agents**; • **Security and privacy** → **Network security**.

Keywords

compound AI systems, LLM agents, hierarchical agent architectures, tool-augmented language models, context engineering, inference-time scaling, token efficiency, cost-performance trade-offs, adversarial POMDP, autonomous cyber defense, catastrophic failures

ACM Reference Format:

Igor Bogdanov, Chung-Horng Lung, Thomas Kunz, Jie Gao, Adrian Taylor, and Marzia Zaman. 2026. Context, Reasoning, and Hierarchy: A Cost-Performance Study of Compound LLM Agent Design in an Adversarial POMDP. In *ACM Conference on AI and Agent Systems (ACM CAIS '26)*, May 26–29, 2026, San Jose, CA, USA. ACM, New York, NY, USA, 29 pages. <https://doi.org/10.1145/3786335.3813149>

1 Introduction

Practitioners build compound LLM agents by composing three design dimensions: *context engineering* [6] (what the agent sees), *deliberation* (reasoning depth), and *hierarchical decomposition* (task distribution). While often assumed to be additive, these choices interact destructively in adversarial, partially observable sequential environments. We present a controlled empirical study of these interactions within a compound LLM agent defending a network in the CybORG CAGE-2 POMDP. By systematically ablating context, reasoning, and hierarchy across five model families and six models,



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            <persName><forename type="first"
            <email>igorbogdanov@cmail.carlet

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source: <https://arxiv.org/abs/2605.16205>

source: xml generated by GROBID



source:

https://es.123rf.com/photo_9865212_pensando-en-hombre-con-burbuja-de-pensamientos-por-encima-de-la-cabeza-ilustraci%C3%B3n-3d.html



source <https://github.com/ollama/ollama>

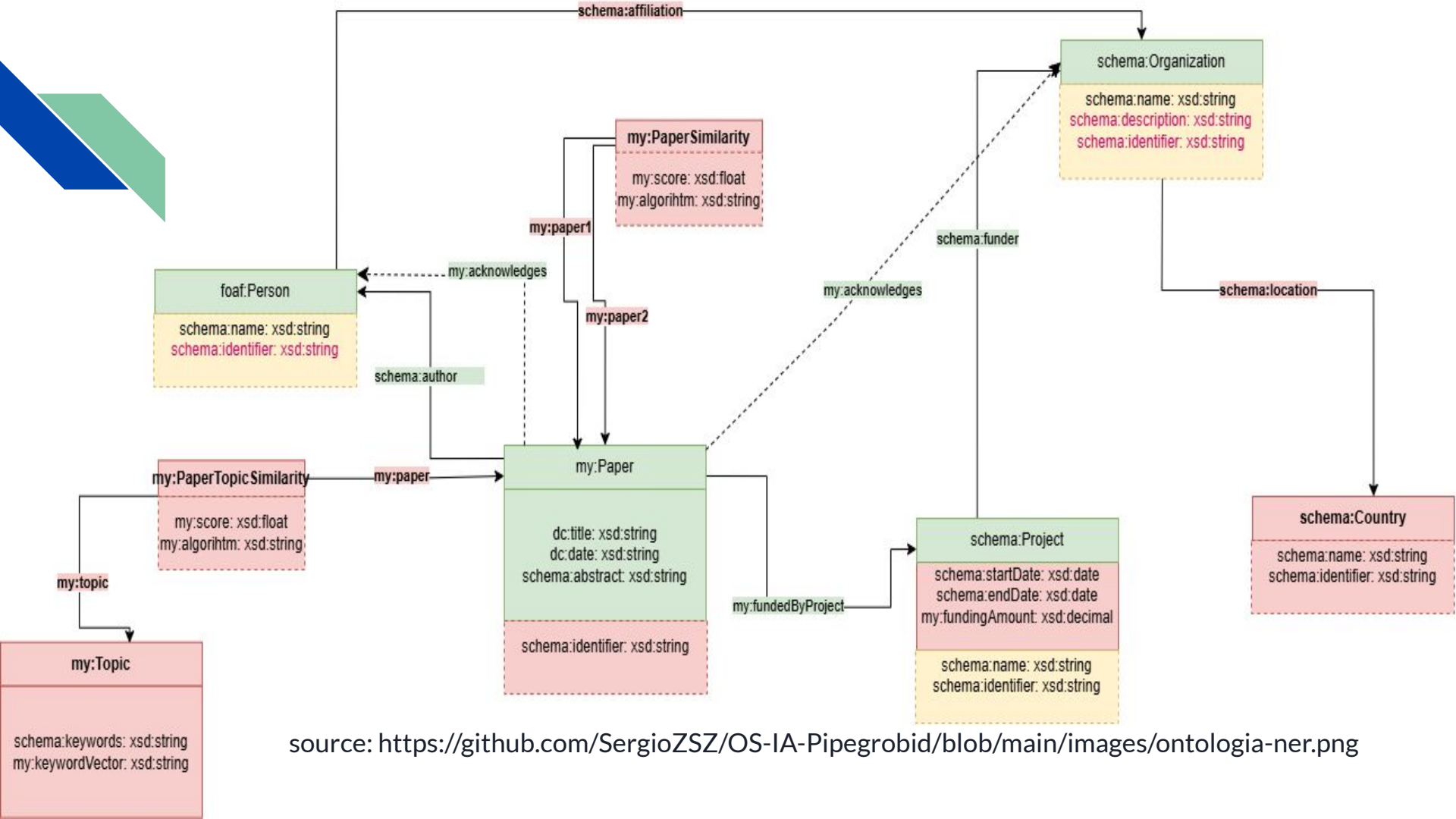
ROBERTA vs KALAWINKA vs Qwen vs LLama

Comparative summary (F1 per category)

Model	PER F1	ORG F1	PROJ F1	Overall F1
groq/llama-3.3-70b-versatile	0.9855	0.9200	1.0000	0.9635
hf/Qwen/Qwen2.5-7B-Instruct	0.7368	0.8511	0.8571	0.8000
Jean-Baptiste/roberta-large-ner-english	0.7143	0.6182	0.0000	0.6269
kalawinka/flair-ner-acknowledgments	0.8254	0.5778	1.0000	0.7619

Source:

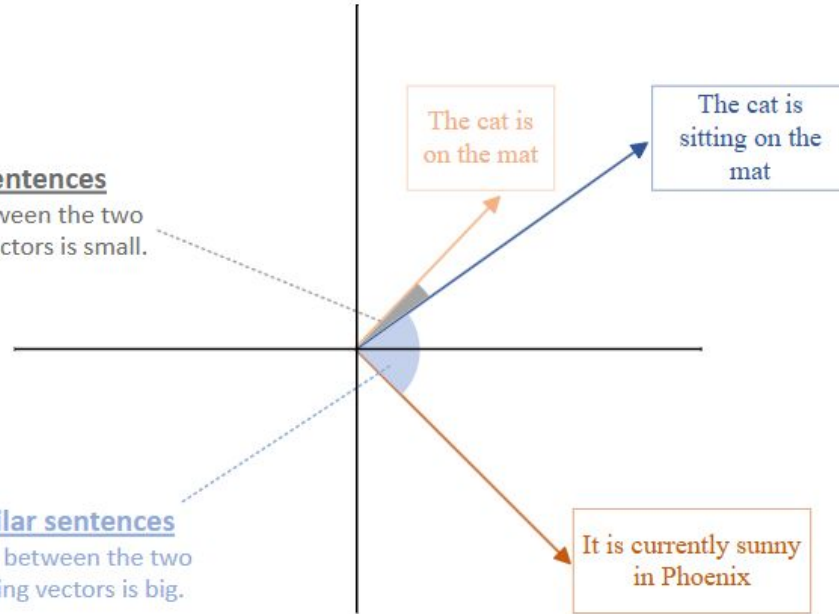
https://github.com/SergioZSZ/OS-IA-Pipegrobid/blob/main/assignment_2/step_2/ner_evaluation/reports/evaluation_report.md



source: <https://github.com/SergioZSZ/OS-IA-Pipegrobid/blob/main/images/ontologia-ner.png>

Similar sentences

The angle between the two embedding vectors is small.



Dissimilar sentences

The angle between the two embedding vectors is big.

BERTopic

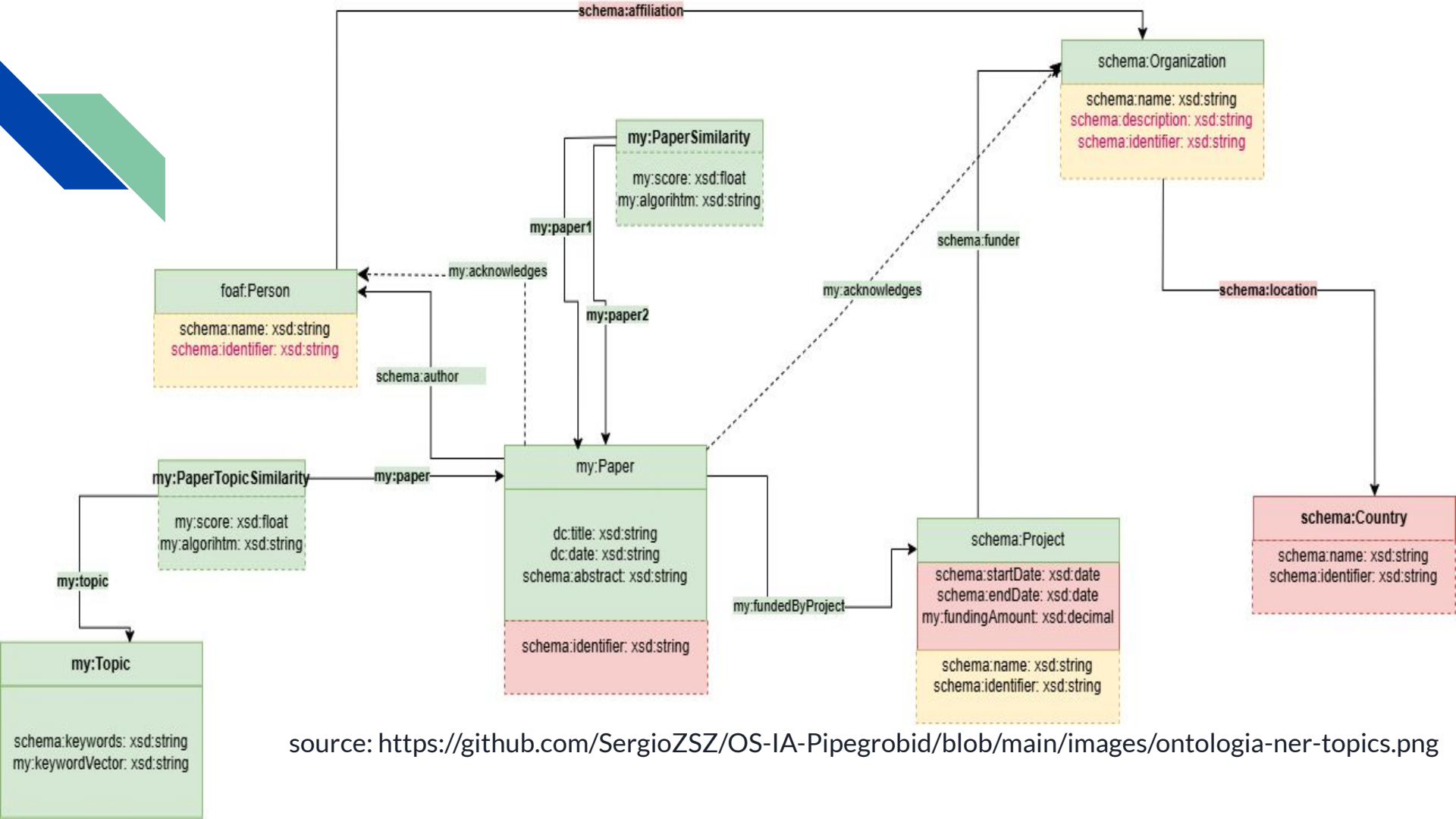


source:

<https://repository-images.githubusercontent.com/297672263/c0a74180-1eb6-11eb-8628-6a14e45344d5>

source:

<https://learn.microsoft.com/es-es/azure/architecture/ai-ml/guide/rag/rag-generate-embeddings>





ORCID
(API REST)

Datos de personas:
identificador único
y afiliación



source: https://es.wikipedia.org/wiki/Archivo:ORCID_logo.svg

Wikidata
(SPARQL)

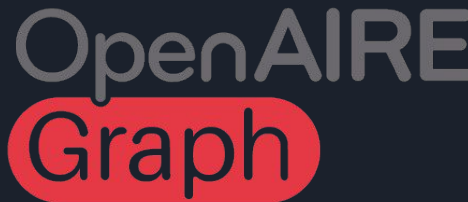
Datos de
organizaciones:
descripción y país



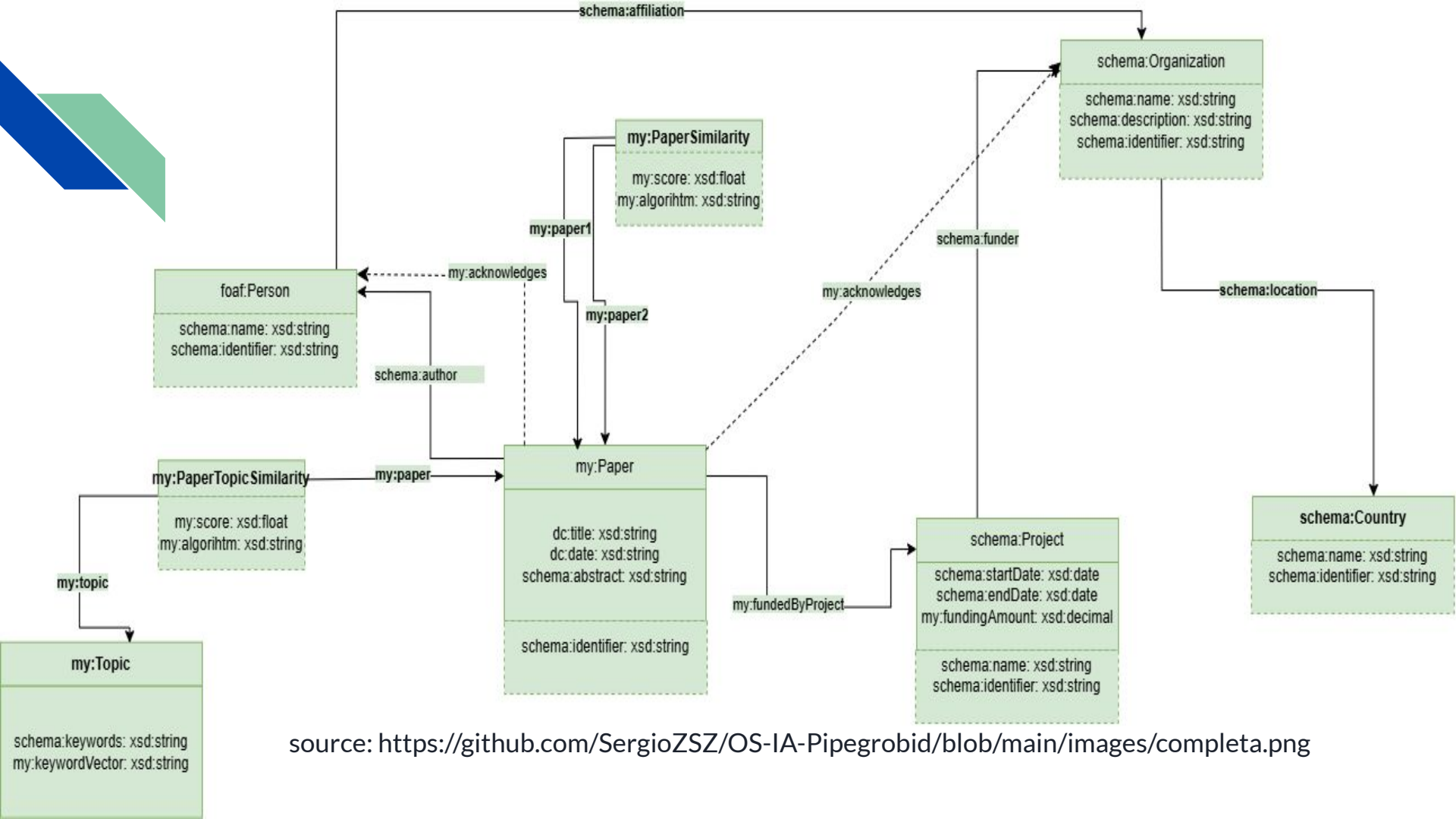
source: <https://es.wikipedia.org/wiki/Wikidata>

OpenAIRE
(API REST)

Datos de proyectos:
título, fechas y
cantidad financiada



source:
<https://connect.geant.org/2025/06/10/openaire-supporting-the-future-of-scholarly-communication-one-e-infrastructure-at-a-time>



source: <https://github.com/SergioZSZ/OS-IA-Pipegrobid/blob/main/images/completa.png>

Funding

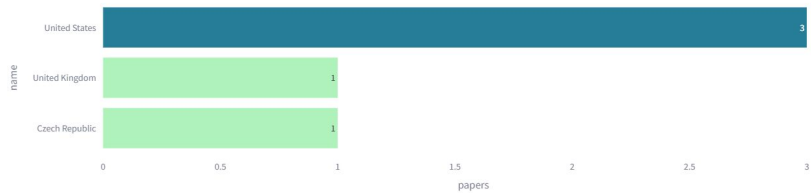
Vista principal: rankings de financiacion filtrables por area tematica.

Filtrar rankings por topic

Todos

Ranking de paises

Top 5 paises por papers financiados



Research Funding Knowledge Graph

Resumen del KG y estado de los servicios principales.

API: ok

KG store: Apache Jena Fuseki

Papers

35

Autores

231

Organizaciones

119

Proyectos

17

Países

9

Topics

4

Similitudes

31

Informacion del backend

Topics

Topics, keywords y papers representativos.

topic_id	name	keywords	papers_count	papers
0	0_reasoning_model_models_ai	reasoning, model, models, ai, human, performance, learning, based, llm, causal	20	paper01 paper02
1	1_dynamics_antimicrobial_household_selection	dynamics, antimicrobial, household, selection, susceptible, size, diversity, infected, b	6	paper14 paper15
2	2_jwst_atmospheric_exoplanet_atmospheres	jwst, atmospheric, exoplanet, atmospheres, retrievals, spectra, planetary, results, ret	5	paper31 paper32
3	3_peptide_protein_instabilities_peptide cyclization	peptide, protein, instabilities, peptide cyclization, cyclization, interface, muco, passiv	4	paper11 paper12

Topic para explorar papers

0_reasoning_model_models_ai

Ver papers de este topic



Referencias:

RFKG: <https://github.com/SergioZSZ/OS-IA-Pipegrobid>

Chowlk: <https://github.com/oeg-upm/Chowlk>

GROBID: <https://github.com/grobidOrg/grobid>

Llama LLM: <https://www.llama.com/>

ORCID: <https://orcid.org/>

Wikidata: <https://www.wikidata.org/>

OpenAIRE Graph: <https://graph.openaire.eu/>

BERTopic: <https://bertopic.com/>

Fuseki: <https://jena.apache.org/documentation/fuseki2/>